

Quadratic Equations

Formulas rendered by code — KaTeX & MathJax, never image-generated

Subject Mathematics

Grade Grade 8

Topic Quadratic Equations

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Lesson Learning Outcomes ★ 🌱

2 min

By the end of the lesson, the learner should be able to:

- a) Recall the standard form of a quadratic equation.
- b) Solve a quadratic equation using the quadratic formula.
- c) Apply related formulas to real-world problems.



Key formulas ★ ★

STANDARD FORM

$$ax^2 + bx + c = 0$$

QUADRATIC FORMULA

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

DISCRIMINANT

$$\Delta = b^2 - 4ac$$

AREA OF A CIRCLE

$$A = \pi r^2$$

$$a^2 + b^2 = c^2$$

★ Same formula via the MathJax engine (SVG) ✿ ✨

QUADRATIC FORMULA — MATHJAX TO SELF-CONTAINED SVG

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

📺 Worked example 🌿 ★



Solve $x^2 - 5x + 6 = 0$. Here $a = 1$, $b = -5$, $c = 6$, so the discriminant is $\Delta = 25 - 24 = 1$, and the quadratic formula gives $x = 2$ or $x = 3$.

✍️ Assessment Questions ★ ✨

5 min

a) Write the quadratic formula from memory.

b) Solve $x^2 - 7x + 12 = 0$.

c) State the discriminant of $2x^2 + 3x - 5 = 0$.

